FREE INDUSTRY GUIDE

ENGINEER'S IT PERFORMANCE GUIDE

Optimise your CAD workstations, secure your IP, and eliminate network bottlenecks

> Engineering firms face unique IT challenges. Large CAD files, legacy CNC equipment, and the need for precision mean you can't afford slow networks or security gaps. It shows you how to stop slow networks, underpowered CAD machines and security gaps costing you contracts and billable hours.

15 MIN	19	DEC 2025
READ TIME	CHECKLIST ITEMS	LAST UPDATED

WHAT'S IN THIS GUIDE

1. CAD Workstation Specifications
2. Network Performance for Large Files
3. Securing Legacy CNC Equipment
4. Cyber Essentials for Supply Chain

CAD WORKSTATION SPECIFICATIONS

Your CAD workstation is the heart of your productivity. Underpowered machines mean waiting for assemblies to load, renders to complete, and files to save.

Minimum Specs for SolidWorks/AutoCAD (2025):

- CPU: Intel Core i7-13700 or AMD Ryzen 7 7700X (8+ cores)
- RAM: 32GB DDR5 (64GB for large assemblies)
- GPU: NVIDIA RTX A2000 or Quadro equivalent (ISV certified)
- Storage: 1TB NVMe SSD + secondary drive for projects
- Display: 27"+ 4K IPS panel

Why ISV Certification Matters:

Graphics cards certified by software vendors (ISV) are tested to work perfectly with your CAD software. Consumer gaming cards may render incorrectly or crash during complex operations.

The Hidden Cost of Consumer Hardware:

A £1,500 gaming PC might seem cheaper than a £2,500 workstation, but:

- No onsite warranty (you ship it away for weeks)
- No ISV certification (random crashes and display glitches)
- Shorter lifespan (not designed for 8+ hours daily use)
- No enterprise management features (Intune, TPM 2.0)

ACTION CHECKLIST

- [ ] **Audit current workstation specs against software requirements** — HIGH PRIORITY
  Tip: Check your CAD vendor's hardware recommendations

- [ ] **Ensure all workstations have 32GB+ RAM** — CRITICAL

- [ ] **Verify graphics cards are ISV certified** — HIGH PRIORITY
  Tip: Check nvidia.com/quadro for certified drivers

- [ ] **Replace mechanical hard drives with NVMe SSDs** — CRITICAL

- [ ] **Confirm workstations have TPM 2.0 for Windows 11** — RECOMMENDED

NETWORK PERFORMANCE FOR LARGE FILES

When your assemblies are 5GB+ and you're working remotely, network performance is critical.

Local Network Optimisation:

- Upgrade to 2.5GbE or 10GbE for file server connections
- Use managed switches with QoS for CAD traffic priority
- Ensure WiFi 6E access points in design offices
- Consider a dedicated VLAN for CAD workstations

Remote Access Without VPN Lag:

Traditional VPNs route ALL traffic through your office, creating bottlenecks. Modern solutions:

- SharePoint/OneDrive for file sync (works offline)
- Azure Files for legacy apps requiring mapped drives
- Parsec or similar for remote workstation access (GPU streaming)

File Server vs Cloud Storage:

For engineering, a hybrid approach works best:

- Active projects: Local NAS with 10GbE connection
- Archive projects: Cloud storage (cheaper, disaster recovery included)
- Automatic tiering: Move inactive files to cloud after 90 days

ACTION CHECKLIST

- [ ] **Test file transfer speeds between workstations and server** — HIGH PRIORITY
  Tip: Should achieve 100MB/s+ for local transfers

- [ ] **Implement QoS rules for CAD application traffic** — RECOMMENDED

- [ ] **Configure OneDrive Files On-Demand for project folders** — HIGH PRIORITY

- [ ] **Set up automatic archival for completed projects** — RECOMMENDED

SECURING LEGACY CNC EQUIPMENT

Many engineering firms have CNC machines running Windows XP or proprietary controllers that cannot be updated. These are security nightmares but essential for production.

The Risk:

Legacy systems with known vulnerabilities are prime ransomware targets. If your CNC machine is on the same network as your email, an infected attachment could encrypt your production line.

The Solution: Network Segmentation

Create isolated VLANs (Virtual LANs) for legacy equipment:

- VLAN 1: Office network (PCs, email, internet)
- VLAN 2: Production network (CNC, PLCs, HMIs)
- Firewall rules: Block internet access for VLAN 2
- Specific allow rules: Only required connections between VLANs

Additional Protections:

- Disable USB ports on legacy machines (prevents infected drives)
- Use application whitelisting where possible
- Monitor network traffic for anomalies
- Document all connections for audit purposes

ACTION CHECKLIST

- [ ] **Identify all legacy systems on the network** — CRITICAL
- [ ] **Create network diagram showing legacy equipment connections** — HIGH PRIORITY
- [ ] **Implement VLAN segmentation for production equipment** — CRITICAL
- [ ] **Block internet access for legacy machines** — CRITICAL
- [ ] **Disable USB ports on CNC controllers** — HIGH PRIORITY

CYBER ESSENTIALS FOR SUPPLY CHAIN

Many Tier 1 manufacturers and defence contractors now require suppliers to hold Cyber Essentials Plus certification. Without it, you may be excluded from tender lists.

Why Engineering Firms Need Certification:

- MOD contracts mandate Cyber Essentials Plus
- Large manufacturers (JLR, Rolls-Royce, Airbus) require it from suppliers
- Insurance premiums are lower for certified businesses
- It proves due diligence if you suffer a breach

The 5 Controls You Must Implement:

1. Firewalls: Secure your internet boundary
2. Secure Configuration: No default passwords, minimal software
3. Access Control: Least privilege, no shared admin accounts
4. Malware Protection: EDR on all endpoints
5. Patch Management: Updates within 14 days

Engineering-Specific Challenges:

- Legacy CNC machines may need to be excluded from scope
- CAD software requires admin rights for some operations
- Remote access for field engineers needs careful configuration

ACTION CHECKLIST

- [ ] **Review supply chain requirements for certification** — HIGH PRIORITY
- [ ] **Conduct gap analysis against Cyber Essentials controls** — CRITICAL
- [ ] **Implement automated patch management** — CRITICAL
- [ ] **Remove admin rights from daily user accounts** — HIGH PRIORITY
- [ ] **Document scope exclusions for legacy equipment** — RECOMMENDED

READY TO IMPLEMENT?

This guide gives you the knowledge. We provide the expertise to make it happen. Book a free consultation to discuss your engineering IT needs.

Contact Fresh Tech for a free consultation:

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